

Technical Notes and Correspondence

Output Feedback Consensus for Networked Heterogeneous Nonlinear Negative-Imaginary Systems With Free-Body Motion

Kanghong Shi [✉], Graduate Student Member, IEEE, Ian R. Petersen [✉], Life Fellow, IEEE, and Igor G. Vladimirov [✉]

Abstract—This article provides a protocol to address the robust output feedback consensus problem for networked heterogeneous nonlinear negative-imaginary (NI) systems with free body dynamics. We extend the definition of nonlinear NI systems to allow for systems with free-body motion. A new stability result is developed for the interconnection of a nonlinear NI system and a nonlinear output strictly negative-imaginary (OSNI) system. Also, a class of networked nonlinear OSNI controllers is proposed to achieve output feedback consensus for heterogeneous networked nonlinear NI systems. We show that in this control framework, the system outputs converge to the same limit trajectory. This consensus protocol is illustrated by a numerical example.

Index Terms—Consensus, free-body motion, heterogeneous systems, nonlinear negative-imaginary (NI) systems, robust control.

I. INTRODUCTION

Negative-imaginary (NI) systems theory was introduced by Lanzon and Petersen in [1] and [2] to address a robust control problem for flexible structures and has attracted much attention among control theory researchers (see [3], [4], [5], [6], [7]). Typical NI systems are systems with colocated force actuators and position sensors. Positive-real (PR) systems theory [8] cannot be applied to the control of such systems in general. An NI system was initially defined in [1] to be a stable system with a frequency response $F(j\omega)$ satisfying $j(F(j\omega) - F(j\omega)^*) \geq 0$ for all $\omega > 0$. Examples of such systems arise in lightly damped structures [9], [10] and nano-positioning systems [11], [12]. The stability of an NI system with transfer function matrix $F(s)$ can be guaranteed by applying a feedback with a strictly negative-imaginary (SNI) controller with transfer function matrix $G(s)$ such that the dc gain condition $\lambda_{\max}(F(0)G(0)) < 1$ is satisfied [1].

The definition of NI systems in [1] and [2] was extended in [7] to include systems with poles in the closed left half of the complex plane except at the origin. The definition has been extended again in [13] to include systems with poles at the origin. Systems with free-body motion, such as single integrators and double integrators were included

in this new definition and a new robust stability result for NI systems was provided in [13]. The original definition of NI systems has also been recently extended to include nonlinear systems in [14] and some stability results were established for nonlinear NI systems in [14] and [15]. However, systems with free-body motion are excluded in the nonlinear NI definition in [14].

Cooperative control for multiagent systems has been a highly active research area over the past two decades [16]. This control paradigm enables multiple systems to perform team missions and has been applied on autonomous vehicles including mobile robots, unmanned air vehicles, autonomous underwater vehicles, and other applications [17]. Consensus, that is convergence to an agreement through information sharing among agents, is one of the most important problems in the area of cooperative control [18]. Consensus algorithms were first studied for first-order dynamics (see [19], [20], etc) and then extended to second-order systems (see [21], [22], [23], etc).

As NI properties arise naturally in autonomous vehicles and a wide variety of other applications [24], consensus problems were investigated in [6] for heterogeneous NI systems. Using NI systems theory, the consensus algorithms in [6] only require outputs to be shared among agents, and consensus is guaranteed if a simple dc gain condition is satisfied. The theoretical results in [6] have already been applied to real-world cooperative control problems (see [25], [26], [27]). However, these results are restricted to linear systems. Motivated by nonlinear NI systems theory, the results of [6] have been recently extended in [28], which provides a protocol that achieves output feedback consensus for networked heterogeneous nonlinear NI systems. However, Shi et al. [28] used the definition of nonlinear NI systems from [14], which excludes systems with free-body motion such as integrators.

NI systems with free-body motion arise in many applications [13], such as rotary cranes [29], flexible link manipulators [30], rotary flexible spacecraft [31], and robotics, which is often of nonlinear nature. For this reason, it is appealing to investigate such systems directly in a nonlinear systems framework. In this work, we introduce a new definition of nonlinear NI systems to include systems with free-body motion. This article provides two main contributions. The first contribution is a stability result for a feedback interconnection of nonlinear NI systems with free-body motion. This result is significant with respect to prior work in the following aspects:

- 1) compared to passivity theory [8], which achieves stability using negative-velocity feedback control, we provide an alternative control scheme, which uses positive-position feedback;
- 2) the works [14], [32] investigated a similar class of nonlinear systems but did not allow for free-body motion, and the assumptions in [14] and [32] indirectly restrict their results to be only applicable to systems that are already asymptotically stable while this article does not have this restriction;

Manuscript received 14 February 2022; revised 7 October 2022; accepted 26 October 2022. Date of publication 14 November 2022; date of current version 30 August 2023. This work was supported by the Australian Research Council under Grant DP190102158. Recommended by Associate Editor S.-J. Chung. (Corresponding author: Kanghong Shi.)

The authors are with the School of Engineering, College of Engineering and Computer Science, Australian National University, Canberra, ACT 2600, Australia (e-mail: kanghong.shi@anu.edu.au; i.r.petersen@gmail.com; igor.g.vladimirov@gmail.com).

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TAC.2022.3221904>.

Digital Object Identifier 10.1109/TAC.2022.3221904

- 3) the paper [13] considered linear NI systems with free-body motion while this article generalizes this result to nonlinear systems.

The second main contribution of this article is in providing a control framework for output feedback consensus of networked heterogeneous nonlinear NI systems with free-body motion. It differs from the prior work in the following aspects:

- 1) it complements passivity-based multiagent control results (e.g., see [33]) by providing a consensus result using NI systems theory;
- 2) the paper [6] only solves a similar problem for linear systems;
- 3) the paper [15] only considers consensus for agents with the same model while this work allows agents to have different system models;
- 4) the papers [15] and [28] only deal with nonlinear NI systems without free-body motion while this restriction is lifted in this work.

To be specific, we obtain a stability result for the interconnection of a nonlinear NI system and a nonlinear output strictly negative-imaginary (OSNI) system (see also [34] and [35] for linear OSNI systems), where nonlinear OSNI systems are also redefined to allow direct feedthrough from the input to the output. Then, we consider the output feedback consensus problem for multiple nonlinear NI systems with different state-space models. We aim to find a control protocol such that the difference between the outputs of any two agents connected in a network converges to zero. We model the communication between these nonlinear NI systems using a weakly connected directed graph, where the nonlinear NI plants and the nonlinear OSNI controllers correspond to the nodes and the edges of the graph, respectively. Each controller takes the difference between the outputs of the two plants connected to it as input and feeds back its output to the plants. This forms an augmented system whose stability is investigated in order to achieve output consensus for the nonlinear NI plants. We prove the networked nonlinear NI plants act as an augmented nonlinear NI system and the networked nonlinear OSNI controllers act as an augmented nonlinear OSNI system. Therefore, the entire networked control system can be regarded as the interconnection of a nonlinear NI system and a nonlinear OSNI system. Similarly to the stability result mentioned above, an extended stability result is established for the networked control system that guarantees output consensus. This shows that for networked heterogeneous nonlinear NI systems satisfying certain assumptions, output feedback consensus can be achieved by using suitable nonlinear OSNI controllers in the proposed control framework. The control framework is robust with respect to bounded perturbations in the system models for both the nonlinear NI plants and the nonlinear OSNI controllers.

The novelty of this work is that the output consensus problem is investigated for nonlinear NI systems with free-body motion for the first time. This provides theoretical support for applications in the fields of formation control, rendezvous, maneuvers, flocking, role assignment, etc. Moreover, from the technical point of view, the innovation of this article is three-fold:

- 1) we defined a class of systems with nonlinear OSNI property to be used as controllers for nonlinear NI plants with free-body motion;
- 2) we constructed novel Lyapunov storage functions used in stability proofs for nonlinear NI systems in single-loop and networked interconnections;
- 3) we proposed a particular network setting to achieve output consensus for nonlinear NI systems with free-body motion.

The rest of this article is organized as follows. Section II defines the notation and provides the required graph theory preliminaries. Section III provides a definition of nonlinear NI systems as well as the stability results for the feedback interconnection of nonlinear NI systems. Section IV provides a control framework for output feedback consensus of networked heterogeneous nonlinear NI systems. A procedure is also provided for the construction of the proposed

control network. In Section V, we use a numerical example to illustrate the procedure of constructing the proposed control network. Finally, Section VI concludes this article.

II. NOTATION AND GRAPH THEORY

Notation: The notation in this article is standard. $\mathbb{R}$ denotes the set of real numbers. $\mathbb{R}^{m \times n}$ denotes the space of real matrices of dimension $m \times n$. A^T and A^* denote the transpose and complex conjugate transpose of a matrix A , respectively. $\lambda_{\max}(\cdot)$ denotes the maximum eigenvalue of a matrix with only real eigenvalues. $A \otimes B$ denotes the Kronecker product of matrices A and B . $(\cdot)$ denotes a constant value for a given vector or scalar signal. $\|\cdot\|$ denotes the Euclidean norm of a vector. I_m is the $m \times m$ identity matrix. $\text{diag}\{a_1, a_2, \dots, a_l\}$ represents a diagonal matrix with the values $a_1, a_2, \dots, a_l$ on its diagonal.

Graph theory preliminaries: $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \{v_1, v_2, \dots, v_N\}$ and $\mathcal{E} = \{e_1, e_2, \dots, e_l\} \subseteq \mathcal{V} \times \mathcal{V}$, describes a directed graph with N nodes and l edges. We consider a directed graph to be weakly connected if the underlying undirected graph obtained by replacing all directed edges of the graph with undirected edges is a connected graph (see, e.g., [17] for the definition of a connected undirected graph). The incidence matrix $Q = [q_{ev}] \in \mathbb{R}^{l \times N}$ of a directed graph is defined so that the elements in Q are given by

$$q_{ev} := \begin{cases} 1 & \text{if } v \text{ is the initial vertex of edge } e \\ -1 & \text{if } v \text{ is the terminal vertex of edge } e \\ 0 & \text{if } v \text{ does not belong to edge } e. \end{cases} \quad (1)$$

III. STABILITY OF A SINGLE-LOOP INTERCONNECTION

In this section, new definitions of nonlinear NI and nonlinear OSNI systems are provided and a new stability result is established for the interconnection of a single nonlinear NI system and a single nonlinear OSNI system. The new stability result is applicable to nonlinear NI systems with free-body motion, which are excluded in the previous stability result given in [15].

Consider the following fairly general nonlinear system:

$$\begin{aligned} \dot{x}(t) &= f(x(t), u(t)) \\ y(t) &= h(x(t)) + Du(t) \end{aligned} \quad (2)$$

where $x(t) \in \mathbb{R}^n$ is the state, $u(t) \in \mathbb{R}^m$ is the input, and $y(t) \in \mathbb{R}^m$ is the output, $f: \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^n$ is a Lipschitz continuous function, $h: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a class C^1 function and $D \in \mathbb{R}^{m \times m}$ is a symmetric matrix; i.e., $D = D^T$.

Definition 1: System (2) is said to be a nonlinear NI system if there exists a positive semidefinite storage function $V: \mathbb{R}^n \rightarrow \mathbb{R}$ of class C^1 such that

$$\dot{V}(x(t)) \leq u(t)^T \tilde{y}(t) \quad (3)$$

for all $t \geq 0$, where

$$\tilde{y}(t) = h(x(t)). \quad (4)$$

In contrast to [14, Definition 3], which excludes linear NI systems with poles at the origin, Definition 1 now includes all linear NI systems satisfying the definition given in [13] by allowing the storage function of the system to be positive semidefinite instead of positive definite.

Definition 2: System (2) is said to be a nonlinear OSNI system if there exists a positive semidefinite storage function $V: \mathbb{R}^n \rightarrow \mathbb{R}$ of class C^1 and a scalar $\epsilon > 0$ such that

$$\dot{V}(x(t)) \leq u(t)^T \tilde{y}(t) - \epsilon \|\tilde{y}(t)\|^2 \quad (5)$$

for all $t \geq 0$, where $\tilde{y}(t)$ is as defined in (4). In this case, we also say that system (2) is nonlinear OSNI with degree of output strictness ϵ .

In this article, nonlinear OSNI systems defined as in Definition 2 are applied as controllers to achieve robust output feedback consensus for networked nonlinear NI plants defined as in Definition 1. First, we provide a stability result for a single feedback interconnection of nonlinear NI systems.

Consider a multiple-input multiple-output (MIMO) nonlinear NI system H_1 with the following state-space model:

$$H_1 : \dot{x}_1(t) = f_1(x_1(t), u_1(t)) \quad (6a)$$

$$y_1(t) = h_1(x_1(t)) \quad (6b)$$

where $x_1(t) \in \mathbb{R}^{n_1}$ is the state, $u_1(t) \in \mathbb{R}^m$ is the input, and $y_1(t) \in \mathbb{R}^m$ is the output, $f_1 : \mathbb{R}^{n_1} \times \mathbb{R}^m \rightarrow \mathbb{R}^{n_1}$ is a Lipschitz continuous function and $h_1 : \mathbb{R}^{n_1} \rightarrow \mathbb{R}^m$ is a class C^1 function.

For the system H_1 with the state-space model (6), we suppose the following assumption is satisfied.

Assumption I: Consider a constant input $u_1(t) \equiv \bar{u}_1$ applied to the system H_1 such that H_1 is in a steady state, i.e., $x_1(t) \equiv \bar{x}_1$ and $y_1(t) = h_1(\bar{x}_1) \equiv \bar{y}_1$ are also constants. For any constant input $\bar{u}_1$ and corresponding constant output $\bar{y}_1$, we suppose that $\bar{u}_1^T \bar{y}_1 \geq 0$.

Assumption I corresponds to a property of all linear NI systems as stated in [1, Lemma 2]. Although this property holds for all linear NI systems, it does not hold for some special nonlinear systems. Hence, we make this assumption in order to exclude these cases.

Consider a MIMO nonlinear OSNI system H_2 with the following state-space model:

$$H_2 : \dot{x}_2(t) = f_2(x_2(t), u_2(t)) \quad (7a)$$

$$y_2(t) = h_2(x_2(t)) + D_2 u_2(t) \quad (7b)$$

where $x_2(t) \in \mathbb{R}^{n_2}$ is the state, $u_2(t) \in \mathbb{R}^m$ is the input, and $y_2(t) \in \mathbb{R}^m$ is the output, $f_2 : \mathbb{R}^{n_2} \times \mathbb{R}^m \rightarrow \mathbb{R}^{n_2}$ is a Lipschitz continuous function, $h_2 : \mathbb{R}^{n_2} \rightarrow \mathbb{R}^m$ is a class C^1 function, and $D_2 \in \mathbb{R}^{m \times m}$ is a symmetric matrix, i.e., $D_2 = D_2^T$.

For the system H_2 with the state-space model (7), we suppose that the following assumption is satisfied.

Assumption II: Consider a constant input $u_2(t) \equiv \bar{u}_2$ applied to the system H_2 such that H_2 is in a steady state, i.e., $x_2(t) \equiv \bar{x}_2$ and $y_2(t) = h_2(\bar{x}_2) \equiv \bar{y}_2$ are also constants. For any constant input $\bar{u}_2$ and corresponding constant output $\bar{y}_2$, we suppose that $\bar{u}_2^T \bar{y}_2 \leq -\gamma \|\bar{u}_2\|^2$ with $\gamma > 0$.

It might be observed that as nonlinear OSNI systems belong to a subclass of nonlinear NI systems, Assumption II seems to be in conflict with Assumption I if the latter was applied to H_2 . In fact, Assumption II can be satisfied because of the term $D_2 u_2(t)$ in the output equation (7b). Note that when stability is considered for feedback controlled or networked nonlinear NI systems, Assumption II only poses a restriction on the choice of controllers. Hence, we can suppose Assumption II, and Assumptions III and IV below hold for the nonlinear OSNI controllers without loss of generality in the nonlinear NI plants to which our results are applicable.

In addition to the abovementioned assumptions, both of the systems H_1 and H_2 are assumed to satisfy the following Assumptions III and IV. For the system H_1 with input $u_1(t)$, state $x_1(t)$, and output $y_1(t) = h_1(x_1(t))$ described by the state-space model (6) and the system H_2 with input $u_2(t)$, state $x_2(t)$, and the auxiliary output $\tilde{y}_2(t) = h_2(x_2(t))$ described by the state-space model (7), we suppose for $i = 1$ and 2, the following conditions are satisfied.

Assumption III: Over any time interval $[t_a, t_b]$ where $t_b > t_a$, $h_i(x_i(t))$ remains constant if and only if $x_i(t)$ remains constant,

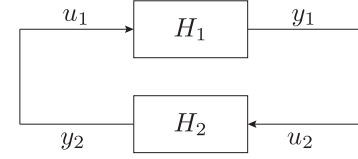


Fig. 1. Closed-loop interconnection of a MIMO nonlinear NI system H_1 and a MIMO nonlinear OSNI system H_2 .

i.e., $\dot{h}_i(x_i(t)) \equiv 0 \iff \dot{x}_i(t) \equiv 0$. Moreover, $h_i(x_i(t)) \equiv 0 \iff x_i(t) \equiv 0$.

Assumption IV: Over any time interval $[t_a, t_b]$ where $t_b > t_a$, $x_i(t)$ remains constant only if $u_i(t)$ remains constant, i.e., $x_i(t) \equiv \bar{x}_i \implies u_i(t) \equiv \bar{u}_i$. Moreover, $x_i(t) \equiv 0 \implies u_i(t) \equiv 0$.

Assumptions III and IV are nonlinear generalizations of standard properties for linear systems. Assumption III is an observability assumption. Assumption IV requires all inputs to be involved in affecting the system dynamics, which can be supposed without loss of generality. In the case of linear systems, Assumption IV corresponds to the matrix B in the realization (A, B, C, D) of the linear system having full column rank.

Theorem 1: Consider the positive feedback interconnection of the system H_1 with state-space model (6) and the system H_2 with state-space model (7), as shown in Fig. 1. Suppose that Assumptions I–IV are satisfied, and the storage function, defined as

$$W(x_1, x_2) := V_1(x_1) + V_2(x_2) - h_1(x_1)^T h_2(x_2) - \frac{1}{2} h_1(x_1)^T D_2 h_1(x_1) \quad (8)$$

is positive definite, where $V_1(x_1)$ and $V_2(x_2)$ are positive semidefinite storage functions that satisfy (3) for the system H_1 and (5) for the system H_2 , respectively. Then, the closed-loop interconnection of the systems H_1 and H_2 is asymptotically stable.

Proof: According to the nonlinear NI property (3) for the system H_1 , the nonlinear OSNI property (5) for the system H_2 and the system setting $u_1(t) \equiv y_2(t)$ and $u_2(t) \equiv y_1(t)$ in Fig. 1, we have

$$\begin{aligned} \dot{V}_1(x_1) &\leq u_1^T \dot{y}_1 = y_2^T \dot{y}_1 = [h_2(x_2) + D_2 u_2]^T \dot{h}_1(x_1) \\ &= [h_2(x_2) + D_2 y_1]^T \dot{h}_1(x_1) \\ &= [h_2(x_2) + D_2 h_1(x_1)]^T \dot{h}_1(x_1) \end{aligned} \quad (9)$$

$$\begin{aligned} \text{and } \dot{V}_2(x_2) &\leq u_2^T \dot{\tilde{y}}_2 - \epsilon \|\dot{\tilde{y}}_2\|^2 = y_1^T \dot{\tilde{y}}_2 - \epsilon \|\dot{\tilde{y}}_2\|^2 \\ &= h_1(x_1)^T \dot{h}_2(x_2) - \epsilon \|\dot{h}_2(x_2)\|^2 \end{aligned} \quad (10)$$

where the abovementioned equalities also use (6b) and (7b). We obtain the time derivative of the storage function $W(x_1, x_2)$ in (8) using (9) and (10)

$$\begin{aligned} \dot{W}(x_1, x_2) &= \dot{V}_1(x_1) + \dot{V}_2(x_2) - \dot{h}_1(x_1)^T h_2(x_2) \\ &\quad - h_1(x_1)^T \dot{h}_2(x_2) - h_1(x_1)^T D_2 \dot{h}_1(x_1) \\ &\leq [h_2(x_2) + D_2 h_1(x_1)]^T \dot{h}_1(x_1) \\ &\quad + h_1(x_1)^T \dot{h}_2(x_2) - \epsilon \|\dot{h}_2(x_2)\|^2 \\ &\quad - \dot{h}_1(x_1)^T h_2(x_2) - h_1(x_1)^T D_2 \dot{h}_1(x_1) \\ &\quad - h_1(x_1)^T D_2 \dot{h}_1(x_1) \\ &= -\epsilon \|\dot{h}_2(x_2)\|^2 \leq 0. \end{aligned} \quad (11)$$

From this it follows that $\dot{W}(x_1, x_2) = 0$ is only possible when $\dot{h}_2(x_2) = 0$. Hence, $\dot{W}(x_1, x_2)$ can remain zero only if $\dot{h}_2(x_2)$ remains zero; i.e., $\dot{W}(x_1, x_2) \equiv 0 \implies \dot{h}_2(x_2(t)) \equiv 0$. According to Assumptions III and IV, $\dot{h}_2(x_2(t)) \equiv 0 \implies \dot{x}_2(t) \equiv 0 \implies u_2(t) \equiv \bar{u}_2$. Hence, the system H_2 is in a steady state. According to the system setting in Fig. 1, $y_1(t) \equiv u_2(t)$. Hence, using Assumptions III and IV, $\dot{y}_1(t) \equiv 0 \implies \dot{x}_1(t) \equiv 0 \implies u_1(t) \equiv \bar{u}_1$. Thus, the system H_1 is also in a steady state. Then, according to Assumption II, we have

$$\bar{u}_2^T \bar{y}_2 \leq -\gamma \|\bar{u}_2\|^2. \quad (12)$$

If $\bar{u}_2 = 0$, then $\bar{u}_2^T \bar{y}_2 = 0$. According to Assumptions III and IV, and the system setting in Fig. 1, $\bar{u}_2 = 0 \implies \bar{y}_1 = 0 \implies \bar{x}_1 = 0 \implies \bar{u}_1 = 0 \implies \bar{y}_2 = 0 \implies \bar{x}_2 = 0$. Hence, in this case, the system is in equilibrium. Otherwise, if $\bar{u}_2 \neq 0$, then from (12), we have

$$\bar{u}_2^T \bar{y}_2 < 0. \quad (13)$$

Also, according to Assumption I, we have

$$\bar{u}_1^T \bar{y}_1 \geq 0. \quad (14)$$

According to the system setting in Fig. 1, we have $\bar{u}_1 = \bar{y}_2$ and $\bar{y}_1 = \bar{u}_2$. Hence, (14) can be rewritten as $\bar{u}_2^T \bar{y}_2 \geq 0$, which contradicts (13). Thus, we can conclude that $\dot{W}(x_1, x_2)$ cannot remain zero unless $x_1 = 0$ and $x_2 = 0$. Thus, according to LaSalle's invariance principle, $W(x_1, x_2)$ will keep decreasing until $W(x_1, x_2) = 0$. Hence, the equilibrium at $(x_1, x_2) = (0, 0)$ of the closed-loop interconnection is asymptotically stable. ■

Remark 1: In Theorem 1, the positive definiteness of the storage function (8) is one of the conditions of the theorem. In the linear case, the dc gain condition implies the positive definiteness of the storage function of the interconnection of an NI system and an SNI system (see, for example, [1]). However, in the nonlinear case, the positive definiteness of (8) does not automatically follow from Assumption II, which is a nonlinear version of the dc gain condition. This is due to the nonlinear nature of the problem and the nonquadratic nature of the storage function (8). However, since the purpose of controller synthesis is to find suitable controllers for given plants, this positive definiteness condition and the assumptions on the controllers pose no restriction on the nonlinear plants to which this theorem can be applied. The theorem is applicable to all nonlinear NI plants given in (6) satisfying Assumptions I, III, and IV, which, as has been explained, are all mild assumptions. A similar remark also applies to Theorem 2.

IV. OUTPUT FEEDBACK CONSENSUS

Consider N heterogeneous nonlinear plants H_{pi} ($i = 1, 2, \dots, N$) described as

$$\begin{aligned} H_{pi}: \quad \dot{x}_{pi}(t) &= f_{pi}(x_{pi}(t), u_{pi}(t)) \\ y_{pi}(t) &= h_{pi}(x_{pi}(t)) \end{aligned} \quad (15)$$

where $x_{pi}(t) \in \mathbb{R}^{n_{pi}}$ is the state, $u_{pi}(t) \in \mathbb{R}^m$ is the input, and $y_{pi}(t) \in \mathbb{R}^m$ is the output, $f_{pi}: \mathbb{R}^{n_{pi}} \times \mathbb{R}^m \rightarrow \mathbb{R}^{n_{pi}}$ are Lipschitz continuous functions and $h_{pi}: \mathbb{R}^{n_{pi}} \rightarrow \mathbb{R}^m$ are class C^1 functions. These systems operate independently in parallel. The subscript “ p ” indicates that this system will play the role of a plant in what follows. Now we give a definition of output feedback consensus for a network of systems H_{pi} ($i = 1, 2, \dots, N$) of the form (15).

Definition 3: A distributed output feedback control law achieves output feedback consensus for a network of systems if $\|y_{pi}(t) - y_{pj}(t)\| \rightarrow 0$ as $t \rightarrow +\infty$, $\forall i, j \in \{1, 2, \dots, N\}$.

To describe the networked plants using a single system model, we combine their states, inputs, and outputs, respectively, as vectors $X_p = [x_{p1}^T, \dots, x_{pN}^T]^T \in \mathbb{R}^{n_p}$, where $n_p = \sum_{i=1}^N n_{pi}$, $U_p =$

$[u_{p1}^T, \dots, u_{pN}^T]^T \in \mathbb{R}^{N_m}$ and $Y_p = [y_{p1}^T, \dots, y_{pN}^T]^T \in \mathbb{R}^{N_m}$. We denote by $\mathcal{H}_p$ the network of nonlinear NI plants consisting of systems H_{pi} ($i = 1, 2, \dots, N$). The system $\mathcal{H}_p$ is formally defined by the following equations:

$$\mathcal{H}_p: \quad \dot{X}_p = \begin{bmatrix} f_{p1}(x_{p1}, u_{p1}) \\ \vdots \\ f_{pN}(x_{pN}, u_{pN}) \end{bmatrix}, \quad Y_p = \begin{bmatrix} h_{p1}(x_{p1}) \\ \vdots \\ h_{pN}(x_{pN}) \end{bmatrix}. \quad (16)$$

Consider a series of heterogeneous nonlinear OSNI controllers H_{ck} ($k = 1, 2, \dots, l$). The controllers have the following state-space models:

$$\begin{aligned} H_{ck}: \quad \dot{x}_{ck}(t) &= f_{ck}(x_{ck}(t), u_{ck}(t)) \\ y_{ck}(t) &= h_{ck}(x_{ck}(t)) + D_{ck} u_{ck}(t) \end{aligned} \quad (17)$$

where $x_{ck}(t) \in \mathbb{R}^{n_{ck}}$ is the state, $u_{ck}(t) \in \mathbb{R}^m$ is the input, and $y_{ck}(t) \in \mathbb{R}^m$ is the output, $f_{ck}: \mathbb{R}^{n_{ck}} \times \mathbb{R}^m \rightarrow \mathbb{R}^{n_{ck}}$ are Lipschitz continuous functions, $h_{ck}: \mathbb{R}^{n_{ck}} \rightarrow \mathbb{R}^m$ are class C^1 functions and $D_{ck} \in \mathbb{R}^{m \times m}$ are symmetric matrices. These systems operate independently in parallel. The subscript “ c ” indicates that this system will play the role of a controller in what follows. We combine the states, inputs, and outputs, respectively, as vectors $X_c = [x_{c1}^T, \dots, x_{cl}^T]^T \in \mathbb{R}^{n_c}$, where $n_c = \sum_{k=1}^l n_{ck}$, $U_c = [u_{c1}^T, \dots, u_{cl}^T]^T \in \mathbb{R}^{l_m}$ and $Y_c = [y_{c1}^T, \dots, y_{cl}^T]^T \in \mathbb{R}^{l_m}$, where

$$\Pi_c = [h_{c1}(x_{c1})^T, \dots, h_{cl}(x_{cl})^T]^T \in \mathbb{R}^{l_m} \quad (18)$$

$$D_c = \text{diag}\{D_{c1}, \dots, D_{cl}\} \in \mathbb{R}^{l_m \times l_m}. \quad (19)$$

We denote by $\mathcal{H}_c$ the network of nonlinear OSNI controllers consisting of systems H_{ck} ($k = 1, 2, \dots, l$). The system $\mathcal{H}_c$ is formally defined by the following equations:

$$\mathcal{H}_c: \quad \dot{X}_c = \begin{bmatrix} f_{c1}(x_{c1}, u_{c1}) \\ \vdots \\ f_{cl}(x_{cl}, u_{cl}) \end{bmatrix}, \quad Y_c = \begin{bmatrix} h_{c1}(x_{c1}) + D_{c1} u_{c1} \\ \vdots \\ h_{cl}(x_{cl}) + D_{cl} u_{cl} \end{bmatrix}. \quad (20)$$

Lemma 1: If the controllers H_{ck} are nonlinear OSNI systems for all $k = 1, 2, \dots, l$, then the networked controller $\mathcal{H}_c$ is also a nonlinear OSNI system.

Proof: For every nonlinear OSNI system H_{ck} , we have a positive semidefinite storage function $V_{ck}(x_{ck})$ and a constant $\epsilon_k > 0$ such that

$$\dot{V}_{ck}(x_{ck}) \leq u_{ck}^T \tilde{y}_{ck} - \epsilon_k \|\tilde{y}_{ck}\|^2 \quad (21)$$

where $\tilde{y}_{ck} = h_{ck}(x_{ck})$ and ϵ_k is the level of output strictness of the system H_{ck} . For the system $\mathcal{H}_c$, we define its storage function V_c as the sum of the storage functions of all the networked controllers, i.e., $V_c := \sum_{k=1}^l V_{ck}(x_{ck})$, which is positive semidefinite. The time derivative of V_c is

$$\begin{aligned} \dot{V}_c &= \sum_{k=1}^l \dot{V}_{ck}(x_{ck}) \leq \sum_{k=1}^l u_{ck}^T \tilde{y}_{ck} - \sum_{k=1}^l \epsilon_k \|\tilde{y}_{ck}\|^2 \\ &\leq \sum_{k=1}^l u_{ck}^T \tilde{y}_{ck} - \epsilon_{\min} \sum_{k=1}^l \|\tilde{y}_{ck}\|^2 \\ &= U_c^T \dot{\Pi}_c - \epsilon_{\min} \|\dot{\Pi}_c\|^2 \end{aligned} \quad (22)$$

where $\epsilon_{\min} = \min\{\epsilon_1, \epsilon_2, \dots, \epsilon_l\} > 0$. Hence, the system $\mathcal{H}_c$ satisfies the definition of a nonlinear OSNI system and $\epsilon_{\min}$ quantifies the level of output strictness of this system.

We model the communication links between the networked nonlinear NI plants H_{pi} ($i = 1, 2, \dots, N$) using a weakly connected directed

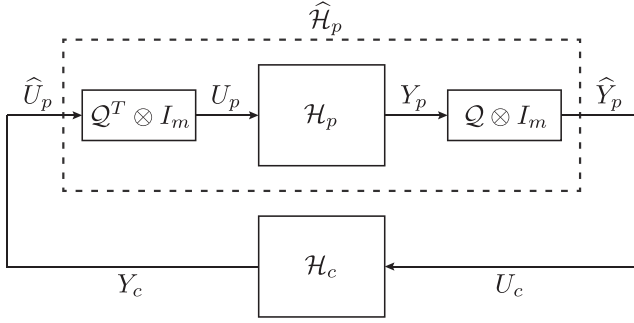


Fig. 2. Positive feedback interconnection of heterogeneous nonlinear NI plants and nonlinear OSNI controllers according to the directed graph network topology.

graph $\mathcal{G}$, which has the corresponding incidence matrix $\mathcal{Q}$ defined by (1). In this article, robust output consensus of heterogeneous nonlinear NI plants is achieved by constructing a control system with the block diagram shown in Fig. 2 and choosing suitable controllers that satisfy certain conditions. The plants and controllers correspond to the nodes and edges of the directed graph, respectively. Each controller takes the difference between the outputs of the two plants connected to it as input. Each plant takes the sum of the outputs of the controllers connected to it as input. Detailed description of the control framework in Fig. 2 can be found in [28], which uses a similar control framework.

As is shown in Fig. 2, the input to the system $\mathcal{H}_c$ in (20) is given by

$$U_c = (\mathcal{Q} \otimes I_m) Y_p. \quad (23)$$

Then, U_p is defined as

$$U_p = (\mathcal{Q}^T \otimes I_m) Y_c \quad (24)$$

where Y_c is the corresponding output of the system $\mathcal{H}_c$ defined in (20). Equivalently, the distributed control protocol for the plant H_{pi} ($i = 1, 2, \dots, N$) is defined by the equations

$$\dot{x}_{ck}(t) = f_{ck}(x_{ck}(t), u_{ck}) \quad (25)$$

$$y_{ck}(t) = h_{ck}(x_{ck}(t)) + D_{ck} u_{ck} \quad (26)$$

$$u_{pi} = \sum_{k=1}^l q_{ki} y_{ck} \quad (27)$$

where q_{ki} is the i th element in the k th row of the incidence matrix $\mathcal{Q}$, as defined in (1). Here, $u_{ck} = \sum_{j=1}^N q_{kj} y_{pj}$ represents the difference between the outputs of the two plants connected by the edge e_k .

The system shown in Fig. 2 can be regarded as the closed-loop interconnection of the networked controller $\mathcal{H}_c$ and the system $\hat{\mathcal{H}}_p$, which is the system contained in the dotted-line block in Fig. 2. This system takes the outputs Y_c of the networked controllers $\mathcal{H}_c$ as input $\hat{U}_p$ and distributes them to each plant via the networked topology. The output $\hat{Y}_p$ of the system $\hat{\mathcal{H}}_p$ is the difference between the outputs for each pair of plants connected by an edge, which is used by the system $\mathcal{H}_c$ as input. We have the following lemma for the system $\hat{\mathcal{H}}_p$.

Lemma 2: If the plants H_{pi} are nonlinear NI systems for all $i = 1, 2, \dots, N$, then the system $\hat{\mathcal{H}}_p$ as shown in Fig. 2 is also a nonlinear NI system.

Proof: According to Definition 1, each nonlinear NI system H_{pi} ($i = 1, 2, \dots, N$) must have a corresponding positive semidefinite

storage function $V_{pi}(x_{pi})$ such that

$$\dot{V}_{pi}(x_{pi}) \leq u_{pi}^T \dot{y}_{pi} \quad (28)$$

where x_{pi} is the state of the system H_{pi} . We define the storage function for the system $\hat{\mathcal{H}}_p$ as $\hat{V}_p = \sum_{i=1}^N V_{pi}(x_{pi})$, which is positive semidefinite. Then

$$\dot{\hat{V}}_p = \sum_{i=1}^N \dot{V}_{pi}(x_{pi}) \leq \sum_{i=1}^N u_{pi}^T \dot{y}_{pi} = U_p^T \dot{Y}_p. \quad (29)$$

According to the system setting in Fig. 2

$$U_p = (\mathcal{Q}^T \otimes I_m) \hat{U}_p, \quad \text{and} \quad \hat{Y}_p = (\mathcal{Q} \otimes I_m) Y_p. \quad (30)$$

Therefore, we have

$$U_p^T Y_p = [(\mathcal{Q}^T \otimes I_m) \hat{U}_p]^T Y_p = \hat{U}_p^T (\mathcal{Q} \otimes I_m) Y_p = \hat{U}_p^T \hat{Y}_p. \quad (31)$$

According to (29) and (31), we obtain the nonlinear NI inequality for the system $\hat{\mathcal{H}}_p$

$$\dot{\hat{V}}_p \leq \hat{U}_p^T \hat{Y}_p. \quad (32)$$

Therefore, $\hat{\mathcal{H}}_p$ is a nonlinear NI system. $\blacksquare$

Before we present the main result, let us make another assumption. For the system $\hat{\mathcal{H}}_p$ in Fig. 2 with input $\hat{U}_p(t)$ and output $\hat{Y}_p(t)$, we suppose the following assumption is satisfied.

Assumption V: Given a constant input $\hat{U}_p(t) \equiv \bar{\hat{U}}_p$ to the system $\hat{\mathcal{H}}_p$, if its output is also constant; i.e., $\hat{Y}_p(t) \equiv \bar{\hat{Y}}_p$, then we suppose $\bar{\hat{U}}_p$ and $\bar{\hat{Y}}_p$ satisfy $\bar{\hat{U}}_p^T \bar{\hat{Y}}_p \geq 0$.

In fact, Assumption I applied to all the plants H_{pi} implies Assumption V in the case that H_{pi} are in a steady state for all $i = 1, 2, \dots, N$. This is because when all the plants satisfy Assumption I and are in a steady state, we have $\bar{U}_p^T \bar{Y}_p \geq 0$ and according to the system setting in Fig. 2 we have $\bar{U}_p^T \bar{Y}_p = [(\mathcal{Q}^T \otimes I_m) \bar{\hat{U}}_p]^T \bar{Y}_p = \bar{\hat{U}}_p^T (\mathcal{Q} \otimes I_m) \bar{Y}_p = \bar{\hat{U}}_p^T \bar{\hat{Y}}_p$ similarly to (31). Hence, $\bar{\hat{U}}_p^T \bar{\hat{Y}}_p \geq 0$. However, Assumption V is made for the networked plants $\hat{\mathcal{H}}_p$ in the following theorem instead of Assumption I for each individual plant because Assumption V also allows for the situation in which the input and output of the system $\hat{\mathcal{H}}_p$ are constant, but the individual plants H_{pi} are not all in a steady state. This situation is possible because the matrix $\mathcal{Q} \otimes I_m$ takes the difference between the outputs of the plants. Under constant inputs, if the plants oscillate with a constant difference between their outputs, then this situation is allowed under Assumption V.

Theorem 2: Consider a weakly connected directed graph $\mathcal{G}$ with the incidence matrix $\mathcal{Q}$ that models the communication links for a network of heterogeneous nonlinear NI systems H_{pi} ($i = 1, 2, \dots, N$) of the form (15). Also, consider the heterogeneous nonlinear OSNI control laws H_{ck} ($k = 1, 2, \dots, l$) of the form (17) for all of the edges. Suppose Assumptions III and IV are satisfied for the plants H_{pi} , Assumptions II, III, and IV are satisfied for the controllers H_{ck} (with $\gamma = \gamma_k$ for the controller H_{ck} in Assumption II) and Assumption V is satisfied by the system $\hat{\mathcal{H}}_p$. Also, suppose the storage function, defined as

$$\hat{W} := \sum_{i=1}^N V_{pi}(x_{pi}) + \sum_{k=1}^l V_{ck}(x_{ck}) - \hat{Y}_p^T \Pi_c - \frac{1}{2} \hat{Y}_p^T D_c \hat{Y}_p \quad (33)$$

is positive definite, where V_{pi} and V_{ck} are the corresponding positive semidefinite storage functions that satisfy (28) for the system $\mathcal{H}_{pi}$ and (21) for the system $\mathcal{H}_{ck}$, respectively. Here, $\hat{Y}_p$ is the output of the system $\hat{\mathcal{H}}_p$. Π_c and D_c are terms in the output Y_c of the system $\mathcal{H}_c$ and are defined in (18) and (19). Then output feedback consensus can

be achieved via the control defined by equations (23), (20), (24), or equivalently (25)–(27) in a distributed manner for each plant H_{pi} , as shown in Fig. 2.

Proof: Define $\hat{V}_p = \sum_{i=1}^N V_{pi}(x_{pi})$ and $V_c := \sum_{k=1}^l V_{ck}(x_{ck})$ as in the proofs of Lemmas 2 and 1, respectively. According to (32), (22) and the system setting $\hat{U}_p \equiv Y_c$ and $U_c \equiv \hat{Y}_p$ shown in Fig. 2, in a similar fashion to inequalities (9)–(11), we have

$$\dot{\hat{V}}_p \leq \hat{U}_p^T \dot{\hat{Y}}_p = Y_c^T \dot{\hat{Y}}_p = \dot{\hat{Y}}_p^T [\Pi_c + D_c U_c] = \dot{\hat{Y}}_p^T [\Pi_c + D_c \hat{Y}_p] \quad (34)$$

$$\text{and } \dot{V}_c \leq U_c^T \dot{\Pi}_c - \epsilon_{\min} \|\dot{\Pi}_c\|^2 = \hat{Y}_p^T \dot{\Pi}_c - \epsilon_{\min} \|\dot{\Pi}_c\|^2. \quad (35)$$

According to (34), (35) and the symmetry of D_c in (19), the time derivative of the storage function $\hat{W}$ satisfies the following inequality:

$$\begin{aligned} \dot{\hat{W}} &= \dot{\hat{V}}_p + \dot{V}_c - \dot{\hat{Y}}_p^T \Pi_c - \hat{Y}_p^T \dot{\Pi}_c - \frac{1}{2} \dot{\hat{Y}}_p^T (D_c + D_c^T) \hat{Y}_p \\ &\leq \dot{\hat{Y}}_p^T [\Pi_c + D_c \hat{Y}_p] + \hat{Y}_p^T \dot{\Pi}_c - \epsilon_{\min} \|\dot{\Pi}_c\|^2 - \dot{\hat{Y}}_p^T \Pi_c \\ &\quad - \hat{Y}_p^T \dot{\Pi}_c - \dot{\hat{Y}}_p^T D_c \hat{Y}_p \\ &\leq \epsilon_{\min} \|\dot{\Pi}_c\|^2 \leq 0. \end{aligned}$$

Hence, the closed-loop system is at least Lyapunov stable. Moreover, $\hat{W} = 0$ can hold only if $\dot{\Pi}_c = 0$. In other words, $\hat{W}$ can remain zero only if $\dot{h}_{ck}(x_{ck}(t))$ remains zero for all $k = 1, 2, \dots, l$. According to Assumptions III and IV, $\dot{h}_{ck}(x_{ck}(t)) \equiv 0 \implies \dot{x}_{ck}(t) \equiv 0 \implies u_{ck}(t) \equiv \bar{u}_{ck}$. Hence, H_{ck} is in a steady state for all $k = 1, 2, \dots, l$. We have $U_c(t) \equiv \bar{U}_c$ and $Y_c(t) \equiv \bar{Y}_c$. According to the system setting in Fig. 2 that $\hat{U}_p(t) \equiv Y_c(t)$ and $U_c(t) \equiv \hat{Y}_p(t)$, we also have $\hat{U}_p(t) \equiv \bar{U}_p$ and $\hat{Y}_p(t) \equiv \bar{Y}_p$. According to Assumption V,

$$\bar{U}_p^T \bar{Y}_p \geq 0. \quad (36)$$

According to Assumption II, we have

$$\bar{U}_c^T \bar{Y}_c = \sum_{k=1}^l \bar{u}_{ck}^T \bar{y}_{ck} \leq - \sum_{k=1}^l \gamma_k \|\bar{u}_{ck}\|^2 \leq -\gamma_{\min} \|\bar{U}_c\|^2$$

where $\gamma_{\min} = \min\{\gamma_1, \gamma_2, \dots, \gamma_l\}$. In the case that $\bar{U}_c \neq 0$

$$\bar{U}_c^T \bar{Y}_c < 0$$

which contradicts (36) because $\bar{U}_c^T \bar{Y}_c = \bar{U}_p^T \bar{Y}_p$. In the case that $\bar{U}_c = 0$, all connected plants have the difference between their system outputs being zero. Hence, output consensus has already been achieved. Otherwise, $\hat{W}$ cannot remain zero. According to LaSalle's invariance principle, $\hat{W}$ will keep decreasing until either $\bar{U}_c = 0$ or $\hat{W} = 0$. Thus, output consensus is achieved in both cases. ■

We provide a step by step procedure Procedure 1, which can be followed in order to construct a controller network as in Fig. 2 to achieve output feedback consensus for given networked nonlinear NI plants.

V. ILLUSTRATIVE EXAMPLE

The example given in the following illustrates the consensus result presented in Section IV. In this example, we aim to synchronize the outputs of nonlinear NI systems connected by a given graph. According to the proposed control framework, we construct a controller network and show that all the assumptions are satisfied. Theorem 2 guarantees

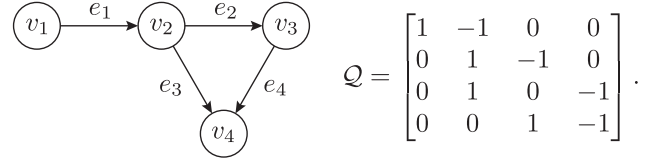


Fig. 3. Weakly connected directed graph $\mathcal{G}$ consisting of four nodes, and its incidence matrix $\mathcal{Q}$.

Procedure 1: Controller Synthesis Algorithm for Output Feedback Consensus of Nonlinear NI Systems.

Given: (a). Networked nonlinear NI plants H_{pi} ($i = 1, \dots, N$) of the form (15); and (b). A graph $\mathcal{G}$ that models the communications between the agents.

Check: (a). All of the plants satisfy Assumptions III and IV; (b). $\mathcal{G}$ is a weakly connected directed graph and the corresponding networked plants $\hat{H}_p$ satisfies Assumption V.

Step 1: For each edge, find a nonlinear OSNI controller H_{ck} of the form (17) such that Assumptions II, III and IV are satisfied.

Step 2: Construct a Lyapunov storage function $\hat{W}$ using the formula given in (33).

Step 3: Adjust the parameters in the controller models to make the Lyapunov storage function $\hat{W}$ positive definite.

Step 4: Apply the control defined by equations (23), (20), (24), or equivalently (25)–(27) in a distributed manner for each plant p_i , as shown in Fig. 2 to achieve output consensus.

output feedback consensus for the plants, which is also verified by a simulation. Consider four nonlinear NI plants H_{pi} at the vertices v_i of the graph in Fig. 3, where the directions of the edges are $e_1 = (v_1, v_2)$, $e_2 = (v_2, v_3)$, $e_3 = (v_2, v_4)$, and $e_4 = (v_3, v_4)$. Then, $\mathcal{Q}$ in Fig. 3 is the incidence matrix of the directed graph $\mathcal{G}$. The plants are nonlinear single integrators which have the following state-space models:

$$\begin{aligned} H_{pi}: \quad \dot{x}_{pi}(t) &= \mu_i u_{pi}(t)^3 \\ y_{pi}(t) &= x_{pi}(t), \quad i = 1, 2, \dots, 4 \end{aligned} \quad (37)$$

where $\mu_1 = 10$, $\mu_2 = 30$, $\mu_3 = 25$, and $\mu_4 = 5$ are constant coefficients. Such plant models involve free-body motion. The storage functions for these four plants are all $V_{pi}(x_{pi}) = 0$. The states x_{p1} , x_{p2} , x_{p3} , and x_{p4} of these four plants have initial values 30, 15, -5 , and -10 , respectively. We aim to synchronise the outputs of these four plants to the same limit trajectory by using nonlinear OSNI controllers H_{ck} at the edges e_k of the graph in Fig. 3

$$\begin{aligned} H_{ck}: \quad \dot{x}_{ck}(t) &= -\alpha_k x_{ck}(t) - \beta_k x_{ck}(t)^3 + u_{ck}(t) \\ y_{ck}(t) &= x_{ck}(t) - u_{ck}(t), \quad k = 1, 2, \dots, 4 \end{aligned} \quad (38)$$

where α_k and β_k are constant coefficients. $\alpha_1 = 5$, $\alpha_2 = 8$, $\alpha_3 = 7$, $\alpha_4 = 3$; $\beta_1 = 3$, $\beta_2 = 2$, $\beta_3 = 5$, and $\beta_4 = 2$, respectively. Controllers 1, 2, 3, and 4 are nonlinear OSNI systems with storage functions $V_{c1} = \frac{5}{2}x_{c1}^2 + \frac{3}{4}x_{c1}^4$, $V_{c2} = 4x_{c2}^2 + \frac{1}{2}x_{c2}^4$, $V_{c3} = \frac{7}{2}x_{c3}^2 + \frac{5}{4}x_{c3}^4$, and $V_{c4} = \frac{3}{2}x_{c4}^2 + \frac{1}{4}x_{c4}^4$, respectively. We now construct the networked plant system $\hat{H}_p$ for the plants (37) using (30) with $m = 1$. Substituting

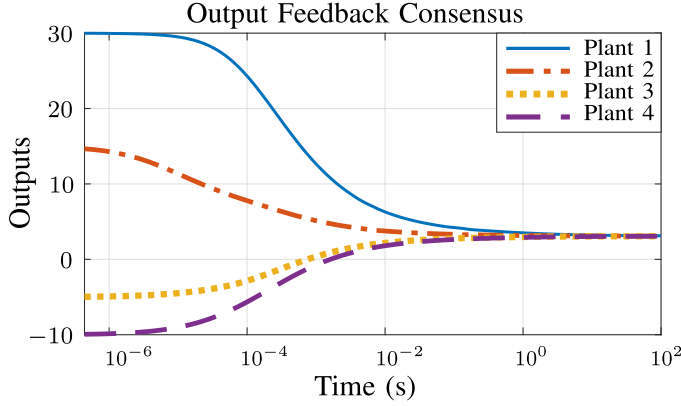


Fig. 4. Outputs of four nonlinear single integrator plants (37), which are synchronized by controllers (38). The figure shows the trajectories of the outputs of the plants from the time that the proposed control law is applied to the time that the outputs converge to the same limit trajectory. Starting from different initial conditions, the outputs of the plants converge to the same limit trajectory under the effect of the proposed control framework in Section IV.

the equations

$$\begin{bmatrix} u_{p1} \\ u_{p2} \\ u_{p3} \\ u_{p4} \end{bmatrix} = Q^T \begin{bmatrix} \hat{u}_{p1} \\ \hat{u}_{p2} \\ \hat{u}_{p3} \\ \hat{u}_{p4} \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} \hat{y}_{p1} \\ \hat{y}_{p2} \\ \hat{y}_{p3} \\ \hat{y}_{p4} \end{bmatrix} = Q \begin{bmatrix} y_{p1} \\ y_{p2} \\ y_{p3} \\ y_{p4} \end{bmatrix}$$

into the system model (37) and letting $\hat{x}_{pi} = \hat{y}_{pi}$, we can obtain the state-space model for the system $\hat{H}_p$ as follows:

$$\begin{aligned} \dot{\hat{x}}_{p1} &= 10\hat{u}_{p1}^3 + 30(\hat{u}_{p1} - \hat{u}_{p2} - \hat{u}_{p3})^3 \\ \dot{\hat{x}}_{p2} &= 30(-\hat{u}_{p1} + \hat{u}_{p2} + \hat{u}_{p3})^3 + 25(\hat{u}_{p2} - \hat{u}_{p4})^3 \\ \dot{\hat{x}}_{p3} &= 30(-\hat{u}_{p1} + \hat{u}_{p2} + \hat{u}_{p3})^3 + 5(\hat{u}_{p3} + \hat{u}_{p4})^3 \\ \dot{\hat{x}}_{p4} &= 25(-\hat{u}_{p2} + \hat{u}_{p4})^3 + 5(\hat{u}_{p3} + \hat{u}_{p4})^3 \\ \hat{y}_{p1} &= \hat{x}_{p1}, \quad \hat{y}_{p2} = \hat{x}_{p2}, \quad \hat{y}_{p3} = \hat{x}_{p3}, \quad \hat{y}_{p4} = \hat{x}_{p4}. \end{aligned}$$

The storage function of the closed-loop system provided by Theorem 2 can be constructed using the formula (33) and also the equations (18) and (19)

$$\begin{aligned} \hat{W} &= \sum_{i=1}^4 V_{pi} + \sum_{k=1}^4 V_{ck} \\ &+ \begin{bmatrix} \hat{x}_{p1} \\ \hat{x}_{p2} \\ \hat{x}_{p3} \\ \hat{x}_{p4} \end{bmatrix}^T \begin{bmatrix} x_{c1} \\ x_{c2} \\ x_{c3} \\ x_{c4} \end{bmatrix} - \frac{1}{2} \begin{bmatrix} \hat{x}_{p1} \\ \hat{x}_{p2} \\ \hat{x}_{p3} \\ \hat{x}_{p4} \end{bmatrix}^T (-I_4) \begin{bmatrix} \hat{x}_{p1} \\ \hat{x}_{p2} \\ \hat{x}_{p3} \\ \hat{x}_{p4} \end{bmatrix} \\ &= \frac{5}{2}x_{c1}^2 + \frac{3}{4}x_{c1}^4 + 4x_{c2}^2 + \frac{1}{2}x_{c2}^4 + \frac{7}{2}x_{c3}^2 + \frac{5}{4}x_{c3}^4 + \frac{3}{2}x_{c4}^2 \\ &+ \frac{1}{2}x_{c4}^4 - \hat{x}_{p1}x_{c1} - \hat{x}_{p2}x_{c2} - \hat{x}_{p3}x_{c3} - \hat{x}_{p4}x_{c4} + \frac{1}{2}\hat{x}_{p1}^2 \\ &+ \frac{1}{2}\hat{x}_{p2}^2 + \frac{1}{2}\hat{x}_{p3}^2 + \frac{1}{2}\hat{x}_{p4}^2 \end{aligned}$$

which is positive definite. Assumptions II–V in Sections III and IV are satisfied. Output feedback consensus is achieved, as illustrated in Fig. 4. Because of the cubic nonlinearity in the plants and the controllers, their outputs have different rates of convergence in the domains where the

states are far and close to their limit values. Therefore, a log scale is used for the time axis in the plot shown in Fig. 4.

VI. CONCLUSION

This article provides a control framework to achieve robust output feedback consensus for networked heterogeneous nonlinear NI systems including systems with free-body motion, using nonlinear OSNI controllers. New definitions for nonlinear NI systems and nonlinear OSNI systems are given and a stability result is established for the simple feedback interconnection of a nonlinear NI plant and a nonlinear OSNI controller. A networked control framework is then considered by modeling the communication topology between systems as a weakly connected directed graph, where the plants are nodes and the controllers are edges. The network of nonlinear NI plants and the network of OSNI controllers are proved to be a nonlinear NI system and a nonlinear OSNI system, respectively. Under reasonable assumptions, output feedback consensus is established for the networked heterogeneous nonlinear NI systems, and the result is robust against variations in the system models of both the plants and controllers provided that the relevant nonlinear NI and nonlinear OSNI properties are preserved. Finally, an example is given to demonstrate the proposed output feedback consensus results.

REFERENCES

- [1] A. Lanzon and I. R. Petersen, "Stability robustness of a feedback interconnection of systems with negative imaginary frequency response," *IEEE Trans. Autom. Control*, vol. 53, no. 4, pp. 1042–1046, May 2008.
- [2] I. R. Petersen and A. Lanzon, "Feedback control of negative-imaginary systems," *IEEE Control Syst. Mag.*, vol. 30, no. 5, pp. 54–72, Oct. 2010.
- [3] Z. Song, A. Lanzon, S. Patra, and I. R. Petersen, "A negative-imaginary lemma without minimality assumptions and robust state-feedback synthesis for uncertain negative-imaginary systems," *Syst. Control Lett.*, vol. 61, no. 12, pp. 1269–1276, 2012.
- [4] J. Xiong, I. R. Petersen, and A. Lanzon, "Finite frequency negative imaginary systems," *IEEE Trans. Autom. Control*, vol. 57, no. 11, pp. 2917–2922, Nov. 2012.
- [5] Z. Song, A. Lanzon, S. Patra, and I. R. Petersen, "Robust performance analysis for uncertain negative-imaginary systems," *Int. J. Robust Nonlinear Control*, vol. 22, no. 3, pp. 262–281, 2012.
- [6] J. Wang, A. Lanzon, and I. R. Petersen, "Robust cooperative control of multiple heterogeneous negative-imaginary systems," *Automatica*, vol. 61, pp. 64–72, 2015.
- [7] J. Xiong, I. R. Petersen, and A. Lanzon, "A negative imaginary lemma and the stability of interconnections of linear negative imaginary systems," *IEEE Trans. Autom. Control*, vol. 55, no. 10, pp. 2342–2347, Oct. 2010.
- [8] B. Brogliato, R. Lozano, B. Maschke, and O. Egeland, "Dissipative systems analysis and control," in *Theory and Applications*, vol. 2. Berlin, Germany: Springer, 2007, pp. 2–5.
- [9] B. Bhikkaji and S. Moheimani, "Fast scanning using piezoelectric tube nanopositioners: A negative imaginary approach," in *Proc. IEEE/ASME Int. Conf. Adv. Intell. Mechatronics*, 2009, pp. 274–279.
- [10] C. Cai and G. Hagen, "Stability analysis for a string of coupled stable subsystems with negative imaginary frequency response," *IEEE Trans. Autom. Control*, vol. 55, no. 8, pp. 1958–1963, Aug. 2010.
- [11] J. van Hulzen, G. Schitter, P. Van den Hof, and J. Van Eijk, "Modal actuation for high bandwidth nano-positioning," in *Proc. Amer. Control Conf.*, 2010, pp. 6525–6530.
- [12] S. Devasia, E. Eleftheriou, and S. R. Moheimani, "A survey of control issues in nanopositioning," *IEEE Trans. Control Syst. Technol.*, vol. 15, no. 5, pp. 802–823, Sep. 2007.
- [13] M. A. Mabrok, A. G. Kallapur, I. R. Petersen, and A. Lanzon, "Generalizing negative imaginary systems theory to include free body dynamics: Control of highly resonant structures with free body motion," *IEEE Trans. Autom. Control*, vol. 59, no. 10, pp. 2692–2707, Oct. 2014.
- [14] A. G. Ghallab, M. A. Mabrok, and I. R. Petersen, "Extending negative imaginary systems theory to nonlinear systems," in *Proc. IEEE Conf. Decis. Control*, 2018, pp. 2348–2353.

- [15] K. Shi, I. G. Vladimirov, and I. R. Petersen, "Robust output feedback consensus for networked identical nonlinear negative-imaginary systems," *IFAC-PapersOnLine*, vol. 54, no. 9, pp. 239–244, 2021.
- [16] R. M. Murray, "Recent research in cooperative control of multivehicle systems," *J. Dyn. Syst., Meas., Control*, vol. 129, no. 5, pp. 571–583, 2007.
- [17] W. Ren, R. W. Beard, and E. M. Atkins, "A survey of consensus problems in multi-agent coordination," in *Proc. Amer. Control Conf.*, 2005, pp. 1859–1864.
- [18] W. Ren, "Consensus strategies for cooperative control of vehicle formations," *IET Control Theory Appl.*, vol. 1, no. 2, pp. 505–512, 2007.
- [19] A. Jadbabaie, J. Lin, and A. S. Morse, "Coordination of groups of mobile autonomous agents using nearest neighbor rules," *IEEE Trans. Autom. Control*, vol. 48, no. 6, pp. 988–1001, Jun. 2003.
- [20] R. Olfati-Saber and R. M. Murray, "Consensus problems in networks of agents with switching topology and time-delays," *IEEE Trans. Autom. Control*, vol. 49, no. 9, pp. 1520–1533, Sep. 2004.
- [21] H. G. Tanner, A. Jadbabaie, and G. J. Pappas, "Stable flocking of mobile agents, Part I: Fixed topology," in *Proc. IEEE 42nd Int. Conf. Decis. Control*, 2003, vol. 2, pp. 2010–2015.
- [22] R. O. Saber and R. M. Murray, "Flocking with obstacle avoidance: Cooperation with limited communication in mobile networks," in *Proc. IEEE 42nd Int. Conf. Decis. Control*, 2003, vol. 2, pp. 2022–2028.
- [23] W. Yu, W. Ren, W. X. Zheng, G. Chen, and J. Lü, "Distributed control gains design for consensus in multi-agent systems with second-order nonlinear dynamics," *Automatica*, vol. 49, no. 7, pp. 2107–2115, 2013.
- [24] I. R. Petersen, "Negative imaginary systems theory and applications," *Annu. Rev. Control*, vol. 42, pp. 309–318, 2016.
- [25] V. P. Tran, M. Garratt, and I. R. Petersen, "Formation control of multi-UAVs using negative-imaginary systems theory," in *Proc. 11th Asian Control Conf.*, 2017, pp. 2031–2036.
- [26] Y. Qi, J. Wang, Q. Jia, and J. Shan, "Cooperative assembling using multiple robotic manipulators," in *Proc. 35th Chin. Control Conf.*, 2016, pp. 7973–7978.
- [27] O. Skeik, J. Hu, F. Arvin, and A. Lanzon, "Cooperative control of integrator negative imaginary systems with application to rendezvous multiple mobile robots," in *Proc. 12th Int. Workshop Robot Motion Control*, 2019, pp. 15–20.
- [28] K. Shi, I. G. Vladimirov, and I. R. Petersen, "Robust output feedback consensus for networked heterogeneous nonlinear negative-imaginary systems," in *Proc. Australian New Zealand Control Conf.*, 2020, pp. 214–219.
- [29] T. Gustafsson, "On the design and implementation of a rotary crane controller," *Eur. J. Control*, vol. 2, no. 2, pp. 166–175, 1996.
- [30] E. Pereira, S. S. Aphale, V. Feliu, and S. R. Moheimani, "Integral resonant control for vibration damping and precise tip-positioning of a single-link flexible manipulator," *IEEE/ASME Trans. Mechatronics*, vol. 16, no. 2, pp. 232–240, Apr. 2011.
- [31] P. C. Hughes, *Spacecraft Attitude Dynamics*. Chelmsford, MA, USA: Courier Corporation, 2012.
- [32] D. Angeli, "Systems with counterclockwise input-output dynamics," *IEEE Trans. Autom. Control*, vol. 51, no. 7, pp. 1130–1143, Jul. 2006.
- [33] N. Chopra and M. W. Spong, "Passivity-based control of multi-agent systems," in *Advances in Robot Control*. Berlin, Germany: Springer, 2006, pp. 107–134.
- [34] P. Bhowmick and S. Patra, "On LTI output strictly negative-imaginary systems," *Syst. Control Lett.*, vol. 100, pp. 32–42, 2017.
- [35] P. Bhowmick and A. Lanzon, "Output strictly negative imaginary systems and its connections to dissipativity theory," in *Proc. IEEE 58th Conf. Decis. Control*, 2019, pp. 6754–6759.